

Ex.No.5: MIMO System Simulation with BER Analysis Date

Objective1:

Program:

```
clc; clear; close all;
```

```
% SNR range
```

```
SNR_dB = 0:2:20;
```

```
SNR = 10.^(SNR_dB/10);
```

```
% MIMO parameters
```

```
Nt = 2; Nr = 2;
```

```
BER = zeros(1,length(SNR));
```

```
for i = 1:length(SNR)
```

```
    bits = randi([0 1],10000,1);
```

```
    ...xlabel('SNR (dB)')
```

```
ylabel('Bit Error Rate (BER)')
```

```
grid on
```

```
% ----- Graph 2: SNR vs Quality Indicator -----
```

```
subplot(2,1,2)
```

```
plot(SNR_dB,1-BER,'-s','LineWidth',2)
```

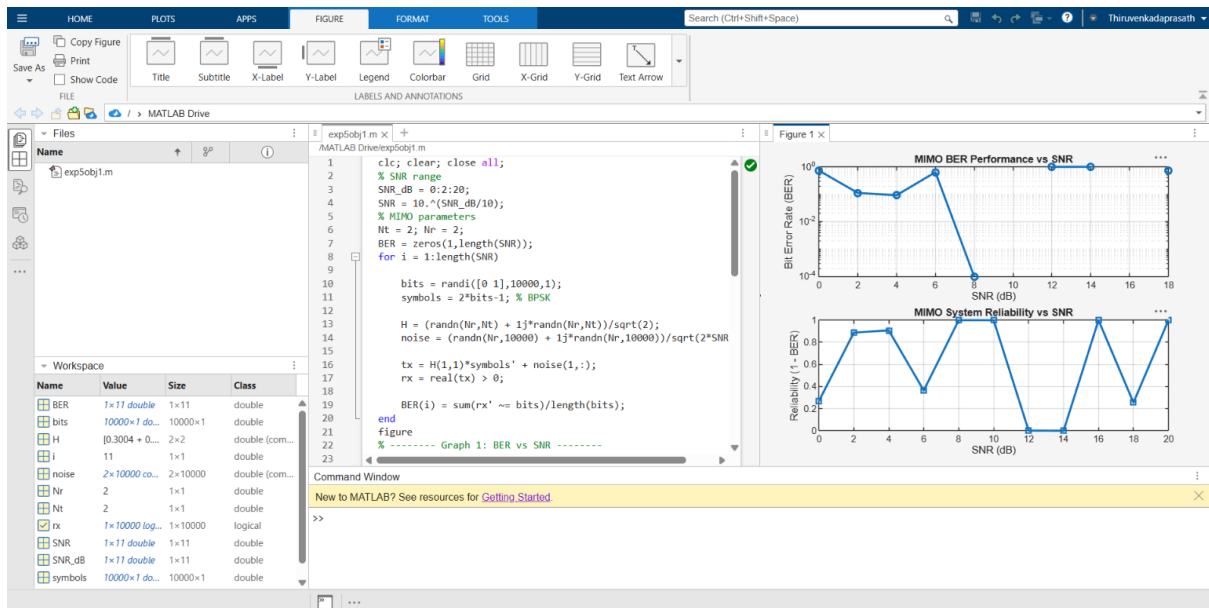
```
title('MIMO System Reliability vs SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('Reliability (1 - BER)')
```

```
grid on
```

Output:



Objective2:

Program:

clc; clear; close all;

% SNR range (dB)

SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO parameters

Nt = 2; Nr = 2;

% Simulated reliability metric (inverse BER model)

Reliability = zeros(1,length(SNR));

for i = 1:length(SNR)

 % simple channel model effect

...xlabel('SNR (dB)')

ylabel('Reliability (0 to 1)')

grid on

% ----- Graph 2: SNR vs Noise Effect -----

subplot(2,1,2)

```
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
```

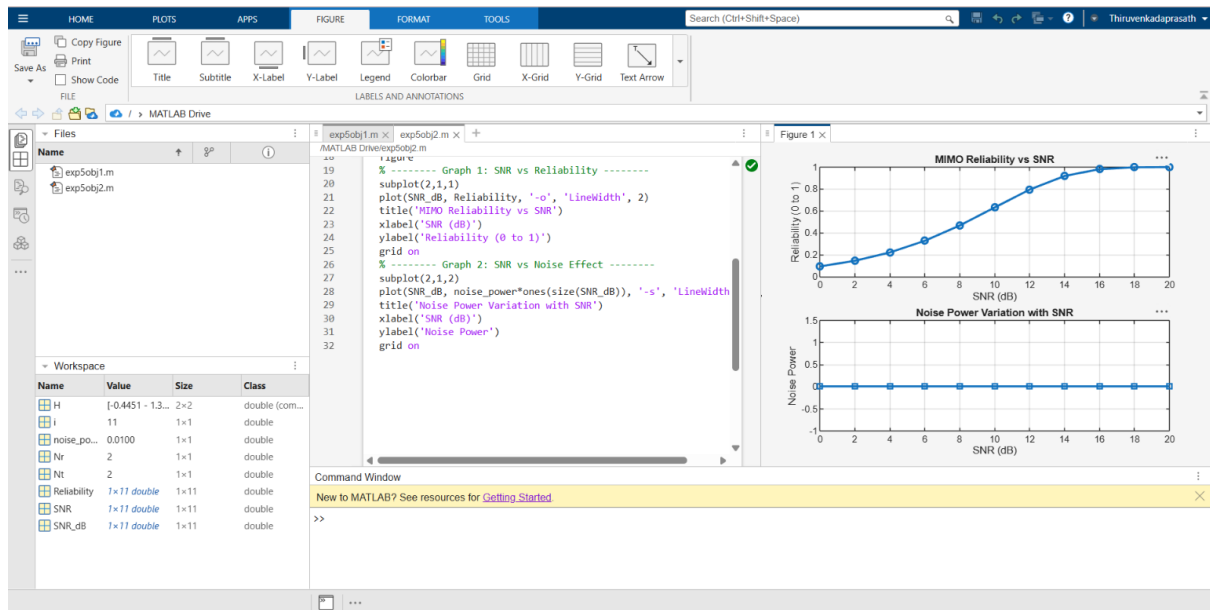
```
title('Noise Power Variation with SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('Noise Power')
```

```
grid on
```

Output:



Objective3:

Program:

```
clc; clear; close all;
```

```
% Resource Block parameters
```

```
rb_bw = 180e3; % 180 kHz per RB
```

```
% Users
```

```
users = 1:5;
```

```
% Allocated Resource Blocks per user
```

```
alloc_RB = [10 12 8 15 5];
```

```
% Modulation efficiency (bits/symbol)
```

```
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
```

% Throughput calculation (Mbps)

...xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2: RB vs Throughput -----

subplot(2,1,2)

plot(alloc_RB, throughput, '-s','LineWidth',2)

title('Resource Blocks vs Throughput')

xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')

grid on

